

Randomization

19/02/2024

Description

This file enables concatenation and randomization of sign strings which occur on different parts of an object (indicated by underscore '_'). We do not know the order of these strings (if any), hence, if we want to combine them, they need to be randomized.

Session Info

Give the session info (reduced).

```
## [1] "R version 3.6.3 (2020-02-29)"
## [1] "x86_64-pc-linux-gnu"
```

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
#library(stringr)
```

Load Data

```
sign.data <- read.csv("~/Github/UpperPalCombinatorics/data/signBase/signBase_clean.csv")
head(sign.data)
```

```
##   object_id ensemble_id      techno_complex      site_name
## 1   sgr0074   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière
## 2   sgr0073   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière
## 3   sgr0072   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière
## 4   sgr0070   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière
## 5   sgr0067   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière
## 6   sgr0062   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière
##               location country longitude latitude  layer
## 1 Saint-Germain-la-Rivière  France   -0.3333    44.4 burial
## 2 Saint-Germain-la-Rivière  France   -0.3333    44.4 burial
## 3 Saint-Germain-la-Rivière  France   -0.3333    44.4 burial
## 4 Saint-Germain-la-Rivière  France   -0.3333    44.4 burial
## 5 Saint-Germain-la-Rivière  France   -0.3333    44.4 burial
## 6 Saint-Germain-la-Rivière  France   -0.3333    44.4 burial
```

```
##          dating_method date_bp_max_min excavation_year          material
## 1 AMS, 858 (Gifa 95456)    15,780 ± 200          1934 animal tooth, red deer
## 2 AMS, 858 (Gifa 95456)    15,780 ± 200          1934 animal tooth, red deer
## 3 AMS, 858 (Gifa 95456)    15,780 ± 200          1934 animal tooth, red deer
## 4 AMS, 858 (Gifa 95456)    15,780 ± 200          1934 animal tooth, red deer
## 5 AMS, 858 (Gifa 95456)    15,780 ± 200          1934 animal tooth, red deer
## 6 AMS, 858 (Gifa 95456)    15,780 ± 200          1934 animal tooth, red deer
##          object_type length_mm width_mm depth_mm preservation
## 1 personal ornament, pendant      NA      NA      NA      complete
## 2 personal ornament, pendant      NA      NA      NA      complete
## 3 personal ornament, pendant      NA      NA      NA      complete
## 4 personal ornament, pendant      NA      NA      NA fragmentary
## 5 personal ornament, pendant      NA      NA      NA fragmentary
## 6 personal ornament, pendant      NA      NA      NA      complete
##  sign_coding coding_certainty coding_clean
## 1      |||X          Certain      |||X
## 2      ||          Certain      ||
## 3      XX      Very uncertain      XX
## 4      ||||          Certain      ||||
## 5      ||      Very uncertain      ||
## 6      |||          Certain      |||
```

Randomize

Randomize sign sequences per object.

```
# set seed to get reproducible result
set.seed(09012020)
# initialize empty vector
coding.random <- c()
# run loop to randomize, and add new column to data frame
for (i in 1:nrow(sign.data)){
  # use single coding per row
  coding.local <- sign.data[i, ]$coding_clean
  # split on underscores surrounded by white spaces
  coding.split <- unlist(strsplit(as.character(coding.local), split = " _ "))
  # randomize order
  coding.random.local <- sample(coding.split)
  # concatenate with underscore and white space again
  coding.random.local <- paste(coding.random.local, collapse = " _ ")
  # add to vector
  coding.random <- c(coding.random, coding.random.local)
}

# add column with randomized order to data frame
sign.data$coding_random <- coding.random

# safe to file
write.csv(sign.data, "~/Github/UpperPalCombinatorics/data/signBase/signBase_randomized.csv", row.names = F)
```